DR-2 via l2-decoupling: a corrected conditional route (separated case rigorous, unrestricted case proof-sketch)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

DR-2 is the unrestricted-class bound $E_+(Q) \le C\, N^{2+\epsilon}$ for EVERY finite $Q \subset S^2$. This note records the ℓ^2-decoupling route to it, corrected after an operator audit. It is NOT a closure: the separated case is rigorous conditional on decoupling; the unrestricted case is a proof sketch. No tier or gate flip is made.

2. Additive energy and a localised L^4 majorant (corrected)

Let $f_Q(x) = \sum_{q \in Q} e^{2\pi i\, q\cdot x}$. The naive torus identity $\int_{[0,1]^3} |f_Q|^4 = E_+(Q)$ is FALSE: it needs $q \in \mathbb{Z}^3$, but $q \in S^2$ is generally non-integer, so the exponential integral is a sinc product, not a Kronecker δ. The correct statements:

$$E_+(Q) = \lim_{R\to\infty} \frac{1}{|B_R|} \int_{B_R} |f_Q(x)|^4\, dx \qquad \text{(Besicovitch mean)},$$

and, for the upper bound that decoupling controls, take a nonnegative Schwartz η with $\widehat{\eta} \ge 0$ everywhere and $\widehat{\eta} \ge 1$ on the ball of radius equal to the minimal nonzero gap of the sumset $Q + Q$; with $\eta_R(x) = R^3 \eta(Rx)$,

$$E_+(Q) \;\le\; C \int_{\mathbb{R}^3} |f_Q(x)|^4\, \eta_R(x)\, dx,$$

because every diagonal quadruple contributes $\widehat{\eta} \ge 1$ and every off-diagonal quadruple contributes $\widehat{\eta} \ge 0$. This localised L^4 integral – not the torus integral – is the object decoupling bounds.

3. Separated case (T6 PROVED CONDITIONAL on decoupling)

Theorem (Bourgain–Demeter 2015). For the sphere/paraboloid in $\mathbb{R}^d$, ℓ^2-decoupling holds for $2 \le p \le p_c = \frac{2(d+1)}{d-1}$; for $d = 3$, $p_c = 4$.

Corollary. If $Q \subset S^2$ is δ-separated with $N \lesssim \delta^{-2}$, apply ℓ^2-decoupling at $p = 4$ to f_Q (one frequency per cap) and bound §2's majorant by the single-cap L^4 norms:

$$E_+(Q) \;\le\; C \int |f_Q|^4 \eta_R \;\lesssim_\epsilon\; N^{2+\epsilon}.$$

This is rigorous modulo the cited theorem – T6 PROVED CONDITIONAL on decoupling.

4. Unrestricted case: multi-scale reduction (T3 PROOF SKETCH)

For arbitrary finite Q (the actual DR-2), decompose by dyadic separation scale $\sigma = 2^{-k}$. At scale σ, points are σ-separated within σ-caps; rescale each cap by σ^{-1} to a unit paraboloid patch (the sphere is locally a paraboloid, height = quadratic), where the separated corollary of §3 applies and gives the intra-scale contribution $\lesssim_\epsilon N_\sigma^{2+\epsilon}$.

OPEN GAP (the load-bearing step). The cross-scale energy – additive quadruples whose four points come from different separation scales – must also be shown $\lesssim_\epsilon N^{2+\epsilon}$, and the $O(\log N)$ dyadic scales summed into the ϵ-loss. This is standard in the decoupling literature (decoupling is intrinsically multi-scale) but is NOT written out here; until it is, the UNRESTRICTED DR-2 is a PROOF SKETCH, not a theorem. The numerics (§6) test exactly this: a clustered cap-grid (the worst non-separated case) still has exponent ≈ 2, not the planar 3.

5. $N^{2+\epsilon}$ is not $N^2 \log^B N$

Decoupling yields $N^{2+\epsilon}$ for every fixed $\epsilon > 0$. This is WEAKER than $N^2 \log^B N$ ($N^2 \log^B N \Rightarrow N^{2+\epsilon}$, not conversely). For the TECT application the admissible mode count n_{adm} is finite (R-009), so N^ϵ is a bounded constant; but the unrestricted ASYMPTOTIC theorem must state $N^{2+\epsilon}$, and the $N^2 \log^B N$ form is a separate (stronger) target not delivered here.

6. Curvature mechanism and numerics (supporting evidence)

The reason the sphere obeys $N^{2+\epsilon}$ while a flat plane allows N^3: an additive quadruple has $|q_1|^2 + |q_2|^2 = |q_3|^2 + |q_4|^2$ on the unit sphere, which with the tangential constraint pins the unordered pair (rectangle rigidity, R-007). R-007 gave the unconditional $N^{5/2}$; decoupling would sharpen it to $N^{2+\epsilon}$. Numerically (dr2_decoupling_exponent.py, 4/4), the empirical growth exponent over $N = 16$–144 is 2.03/1.95/2.01 for great-circle / sphere-cap-grid / random, versus 2.98 for the SAME grid kept FLAT – a direct demonstration that curvature pins the energy from N^3 to N^2. This is strong supporting evidence, not a proof.

7. TECT consequence (none yet)

No consequence is drawn. IF the unrestricted case is completed (the §4 cross-scale step written), DR-2 would become T6 PROVED CONDITIONAL on decoupling, enabling the unrestricted STEP-5B closure and the removal of H-ADM-COH from B1 – but that is explicitly NOT done here. H-ADM-COH STAYS in the B1 hypothesis set; DR2-SHARE stays OPEN; B5 stays T5; B1 stays T6.

8. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"The L^4 = additive-energy identity was wrong."* UPHELD (operator audit): the torus identity needs integer frequencies; corrected to the Besicovitch limit / Schwartz majorant in §2. The first-draft Section 2 was not proof-grade; this is the fix.

(β) *"The unrestricted multi-scale reduction is asserted, not proved."* UPHELD: §4 now writes the reduction but marks the cross-scale energy step OPEN. Hence the unrestricted DR-2 is T3 PROOF SKETCH, not T6; the gate is not flipped.

(γ) *"$N^{2+\epsilon}$ was conflated with $N^2 \log^B N$."* UPHELD: §5 distinguishes them; the route delivers only $N^{2+\epsilon}$.

Quantitative sanity check (mandatory): magnitude – sphere exponents 2.03/1.95/2.01 $\leq$ 2.3, flat control 2.98 $\geq$ 2.6; sign-direction – all energies positive, sphere below flat; limit-case – $d = 3 \Rightarrow p_c = 4$, the additive-energy exponent (critical, the borderline case decoupling reaches); consistency – the cap-grid (worst non-separated config) shows $\approx N^2$, supporting the §4 sketch but not replacing its missing cross-scale step.

9. Result footer

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Result ID:                B5 / DR-2 via l2-decoupling (corrected conditional route)
Precise statement:        SEPARATED Q in S^2: E_+(Q) <=_eps N^{2+eps} is T6 PROVED
                          CONDITIONAL on Bourgain-Demeter decoupling (d=3, p=4),
                          via the Schwartz-majorant localisation (NOT the false
                          torus identity). ARBITRARY finite Q (DR-2 proper): T3
                          PROOF SKETCH -- multi-scale reduction written, cross-scale
                          energy step marked OPEN. Bound is N^{2+eps}, not
                          N^2 log^B N. Numerics: exponent ~2 sphere vs ~3 flat.
Scope:                    Unrestricted-class sphere additive energy; DR2-SHARE.
Dependencies:             Bourgain-Demeter decoupling (BLACK BOX); R-007 (rectangle),
                          R-008 (dyadic), R-021 (cross-energy); operator audit
                          reviews/2026-06-08-dr2-decoupling-critical-audit.md.
Evidence grade:           Separated T6 PROVED CONDITIONAL; unrestricted T3 PROOF
                          SKETCH; + EXECUTED numerics (dr2_decoupling_exponent.py
                          4/4).
Reproduction command:     python codes/vacuum/dr2_decoupling_exponent.py
Expected output:          4/4 PASS; alpha 2.03/1.95/2.01 (sphere) vs 2.98 (flat).
Falsification gate:       A finite Q in S^2 with E_+(Q) >= N^{2.3} (none found; the
                          flat-plane N^3 is excluded by curvature).
Tier before / after:      DR-2: OPEN -> OPEN (NO flip). Separated sub-case T6
                          conditional; unrestricted T3 sketch. B5 T5, B1 T6
                          unchanged; H-ADM-COH STAYS in B1. R-022 downgraded T6->T4.
No-overclaim statement:   This is NOT a DR-2 closure. The first draft over-claimed
                          T6 PROVED CONDITIONAL on a false L^4 identity; corrected
                          here. Decoupling is invoked not reproved; the unrestricted
                          case lacks the cross-scale step. No gate/tier/hypothesis
                          action is taken.
Next required action:     Write the \S4 cross-scale energy bound (close the multi-
                          scale reduction); then DR-2 = T6 PROVED CONDITIONAL on
                          decoupling and operator may consider the DR2-SHARE flip.
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